Ellie Kung

713-269-2150 | ekung@olin.edu | linkedin.com/in/ellie-kung | elliedbq.github.io

Education

Olin College of Engineering Needham, MA
Bachelor of Science in Mechanical Engineering *Aug. 2024 – May 2028*

Experience

Mechanical Engineering Intern June 2026 – August 2026
change:Water Labs *Somerville, MA*

- Redesigned a compact, water resistant, toilet cover/seat actuation mechanism, replacing a linear actuator ramp system with a servo-driven design to address insufficient force and friction losses.
- Developed Arduino code to allow for sensor based actuation of toilet cover and to automate fatigue testing, evaluating the servo for over 100 cycles.
- Integrated an off the shelf push-lock mechanism to hold a toilet cover in place, reducing reliance on continuous servo actuation.
- Developed and evaporation tested an origami inspired membrane cover used to protect the membrane against damage.

Mechanical Engineer Sep. 2026 – Present
Olin Agriculture Robotics Research Team *Needham, MA*

- Designing a 2-axis gantry attachment for an autonomous agricultural vehicle (Amiga) to position a soil probe for automated moisture mapping, with 40 in. of y-axis travel and 20 in. of z-axis stroke.
- Performing first-principles calculations to size drive components, evaluate motor torque, decide on Acme lead screw, and verify rack and pinion requirements under expected loading.

Chassis Lead Engineer May 2025 – June 2026
Olin Baja SAE (Building an off-road car) *Needham, MA*

- Organized onboarding for 10+ new members, improving retention and CAD proficiency in OnShape
- Designed and manufactured roll cage and safety-critical structures, compliant with SAE standards, and jigs for tab welding.

Projects

Robotic Snake | *Onshape, 3d Printing, Arduino, Origami* Oct. 2025 - Dec. 2025

- Successfully designed and coded a six degree of freedom snake that uses bio-mimicry (snake-like movements) to traverse its environment
- Manufactured a waterproof skin based on origami pattern design iteration to protect inner components
- Lead a team of 4 through 3 sprint cycles and design reviews, coordinating design, fabrication, and integration.

Pneumatic T-Shirt Arm (PTA) | *MATLAB, CAD, Mechanical Design* Oct. 2026 - Present

- Collaborating with a three person team to design a portable, pneumatic T-shirt launcher using compressed air to propel a shirt directly through the barrel, triggered by a first pump.
- Evaluating compressed air requirements and portable tank constraints to target room-scale launches while accounting for safe operation.
- Developing an inertia-based trigger mechanism through benchtop models and CAD, using MATLAB calculations to investigate the relationship between trigger rod mass, actuation requirements, and launch-system pressure.
- Conducting preliminary safety testing by measuring velocity of a manually thrown t-shirt.

Precision Machining Projects | *Manual Milling, Turning, GD&T* Sept. 2026 - Present

- Machined precision components, including a 1-2-3 block, machinist's hammer, and clamp, using manual milling and turning processes.
- Manufactured and inspected components to drawing specifications, applying GD&T principles and precision measuring techniques.

Technical Skills

Fabrication: Manual Lathe/Mill, CNC Lathe, CNC Mill, Laser Cutting, 3d Printing, Woodworking, Rapid Prototyping
Languages: Python, MATLAB, C++/Arduino, JavaScript, HTML/CSS
Tools & Platforms: OnShape, SolidWorks, ROS2, Gazebo, RViz, Git, Ubuntu, Raspberry Pi, Microsoft Office